Analog PID Line Follower - Test and Setup Guide

A step-by-step bring-up procedure for the board: what to measure, what you should see, and how to set the offset, gain and PID pots. Values marked typical are expected ranges for this design, not guarantees. Write your real readings in the log at the end.

What you need

- Digital multimeter (DMM)
- Bench supply with current limit (12 V). A battery works only after the board passes the checks below.
- Oscilloscope (strongly recommended for the triangle and PWM)
- White paper or board, black electrical tape (the line), a ruler
- Small screwdriver for the trimmers

Test points on the board: VREF_0, VBASE_1, VERROR_0, P_0, I_1, D_1, PID_0, COM_A1, COM_B1 (duty voltages), PWM_A1, PWM_B1, TRI1.

Order of work: power, then triangle and PWM with motors off, then sensors, then error and offset, then P/D, then motors, then the track. Do not skip ahead. Each stage depends on the one before.

Stage 0 - Before power (motors disconnected)

1. Check polarity of C1-C5, D1, D2, U9 and the MOSFETs Q1, Q2 (pad 1 = gate, pad 3 = drain).
2. Check every IC for pin 1 orientation, and that the op-amps are the rail-to-rail part you chose.
3. With the DMM in resistance mode, measure +12 V to GND and +5 V to GND. Neither should read near 0 ohm. Hundreds of ohms to kilohms is normal.
4. Set the trimmers: OFF_0 at mid, GAIN_0 at minimum, KP_0 at about 10k, KD_0 at minimum, BASE_0 at mid.
5. Leave R302 unfitted (the integrator stays off for now).

Stage 1 - Power

Set the supply to 12 V with a 200 mA current limit.

Check	Where	Expect (typical)	If wrong
Supply current	bench supply	80 - 120 mA (ten LEDs about 17 mA each at 5 V)	Above 200 mA: switch off, look for a short or a hot part
5 V rail	+5 V	4.9 - 5.1 V	Wrong or zero: check U9 pins, L1 value and orientation, CD2
5 V ripple	scope on +5 V, AC coupled	under about 50 mV peak to peak	Too high: check C4/C5 are real ceramics
VREF	VREF_0	2.45 - 2.55 V	Zero: R100/R200 or buffer; unstable: add the 1 uF at the divider midpoint
LED current	across one 220 ohm LED resistor	3.5 - 4 V, so 16 - 18 mA	Much lower: LED reversed or open
Each IC supply	pin 4 and pin 11 of op-amps	5 V between them	Missing: solder or trace

If any of this fails, stop and fix it before continuing.

Stage 2 - Triangle and PWM (motors still off)

Triangle (TRI1)

Quantity	Expect (typical)
Shape	triangle, centered near 2.5 V
Low and high peaks	about 1.7 V and 3.3 V (1.65 V peak to peak)
Frequency	4.5 - 5.2 kHz (calculated 5.05 kHz; slew rate lowers it a little)

If it does not oscillate: check the Schmitt trigger (R306, R307), the integrator (R305, C301) and VREF on the + input.

Base speed (VBASE_1)

Turn BASE_0 end to end. VBASE should sweep about 1.2 V to 3.8 V (R316 and R317 set the ends). With the PID at zero, duty depends on VBASE like this:

VBASE	Duty on PWM_A1 / PWM_B1	Average motor voltage at 12 V
1.7 V	about 2%	0.2 V
2.0 V	20%	2.4 V
2.2 V	32%	3.8 V
2.5 V	50%	6 V
2.8 V	68%	8.2 V
3.0 V	80%	9.6 V
3.3 V	98%	11.8 V

The formula is duty = (VBASE - 1.675) / 1.65. At VBASE of 2.5 V both channels must show the same duty within about 2%. COM_A1 and COM_B1 should both read 5 V minus VBASE (2.5 V at mid). The gate swing at Q1 and Q2 should be about 0 - 5 V with a slightly slow rising edge (a few microseconds, from the 4.7k pull-up).

Stage 3 - Sensors

Hold the board 3 - 5 mm above the surface. This is the most important setting for the sensors. Measure at the op-amp outputs (nets L1 - L5, R1 - R5).

1 Over plain white, write down all ten voltages.

2 Over black tape, write down all ten again.

Surface	Expect (typical for QRE1113 with 4.7k pull-up)
White	about 0.3 - 1.5 V (phototransistor conducting)
Black	about 3.5 - 4.9 V

Because I have not measured your sensors, treat these as a guide. What matters is that the difference between black and white is large (a swing of 2 V or more) and similar on all ten.

Mirror check: for each right-side channel, V(Ri) + V(Rii) must equal 5.0 V within about 0.05 V (R1 with R1i, and so on). This proves the inverting stages work.

Mark the sensors in order S1 (far left) to S10 (far right). S1 and S10 must have the 20k input resistors (R402, R407), and S5 and S6 the 100k (R406, R411).

Stage 4 - Error stage, offset and gain

VERROR_0 is the error voltage. With no error it sits at VREF (2.5 V).

Set the offset

1 Put the board over plain white so all sensors see the same thing. Keep the height fixed.

2 Measure the difference between VERROR_0 and VREF_0 with the DMM.

3 Turn OFF_0 until the difference is 0 mV (aim for under 5 mV).

How far the trim can go depends on the total feedback resistance (R401 plus GAIN_0). The offset reaches about plus or minus 2.5 V times that resistance divided by 1M:

R401 + GAIN_0	Offset trim range at VERROR
10k	plus or minus 25 mV
22k	plus or minus 55 mV
110k	plus or minus 275 mV

The range is small on purpose. If you cannot reach zero, your sensors do not read the same on white. Example: if S1 reads 0.2 V higher than S10 on white, the error moves by about 0.2 V times (feedback resistance divided by 20k), which is 0.22 V at 22k. Fix this by checking sensor height and angle, cleaning the lenses, or replacing a bad sensor. Do this before any tuning.

When you slide the board sideways, VERROR must change sign as you cross the center. Repeat this check again after the offset is set.

Set the gain (GAIN_0)

Gain for each sensor pair is feedback resistance divided by its input resistor, so S1 and S10 have the highest gain (R divided by 20k).

1 Start with GAIN_0 at minimum (total about 10k, outer gain 0.5).

2 Put the board so only S1 sees black, the others white. Watch VERROR_0.

3 Raise GAIN_0 until VERROR is about 1.5 - 2 V away from 2.5 V, but never reaching the rails (stay between about 0.5 V and 4.5 V).

Starting guide: total feedback in kilohms is about 20 x (1.8 / sensor swing in volts). A 3.6 V swing gives 10k, a 1.8 V swing gives 20k, a 1.2 V swing gives 30k. Do the same on S10; the result should be equal and opposite.

Stage 5 - P, D and the direction check

With a line under the left sensors, expect these signs:

Test point	Line on the left	Line on the right
VERROR_0	below 2.5 V	above 2.5 V
P_0	above 2.5 V	below 2.5 V
PID_0	below 2.5 V	above 2.5 V
COM_A1 (duty A)	above 2.5 V, so motor A slows	below 2.5 V
COM_B1 (duty B)	below 2.5 V, so motor B speeds up	above 2.5 V

So the motor driven by Q1 (A) must be your left motor. If your robot later turns away from the line, swap the two motor leads or swap which motor is A and B.

P stage

P_0 moves from 2.5 V by Kp times the VERROR change. Kp equals KP_0 divided by 10k, so 10k gives 1 and 100k gives 10. Check that P_0 swings farther as you turn the pot, and that it does not hit a rail with a normal error (0.2 - 0.5 V).

D stage

D_1 mostly sits at 2.5 V and only twitches when the error changes. With KD_0 at maximum (time constant 1 ms), moving the line across a sensor quickly (a change of about 1 V in 20 ms) should give a deviation of tens of millivolts. With KD_0 at 0 it stays flat. Check the filter capacitor CD3 makes the trace smooth, not noisy.

I stage

I_1 will drift toward a rail with a small constant error, and that is normal while R302 is unfitted. The integrator does not reach the output. Only fit R302 after the PD tuning is good.

PID_0

With zero error PID_0 equals VREF (2.5 V). It should follow the error sign above.

Stage 6 - Motors

Connect the motors with the wheels off the ground. Use a 1 A current limit at 12 V.

1. With BASE_0 at mid (VBASE 2.5 V) both motors should turn at about the same speed. A DMM in DC mode across a motor reads the average, so about 6 V.
2. No-load current of a small 12 V N20 motor is typically 50 - 120 mA each. Much higher means a mechanical or wiring problem.
3. Check Q1 and Q2 stay barely warm. If hot, check gate drive (0 - 5 V) and the 10 ohm gate resistors.
4. Slide a black strip under the left sensors: the left motor slows and the right speeds up. Reverse it: the opposite.
5. Look at the motor terminals on the scope: you should see a PWM waveform near the supply voltage with small spikes clamped by the SS14 diodes.

Stage 7 - On the track

Use a line 15 - 20 mm wide, with gentle curves first. Start from these settings, which are guesses, not known-good values:

- VBASE about 2.2 V (about 32% duty)
- GAIN_0 as set in Stage 4
- KP_0 about 10k - 20k (Kp 1 - 2)
- KD_0 at 0

Tune in this order:

1. Raise Kp slowly until the robot follows the line but wobbles from side to side. Back it off to about 60% of that value.
2. Raise KD_0 a little at a time until the wobble calms down. Too much Kd makes it twitchy and noisy.
3. Raise base speed slowly. If it starts losing the line, lower speed or raise Kp and Kd together.
4. Optional: fit R302 for the integral term only if the robot always leans to one side on long straights. Start with the KI_0 pot at its highest resistance (smallest Ki).

Troubleshooting

Symptom	Likely cause
VERROR stuck at a rail	GAIN_0 too high, or a sensor reading the wrong way (check Stage 3 values)
Robot always turns one way	Offset not set, or one sensor reading differently on white
Turns away from the line	Motors swapped, or the weights are reversed (S1 and S10 must be 20k)
Oscillates on straights	Kp too high, or Kd too low or noisy
Motors do not move	Check PWM_A1, PWM_B1 first, then gate voltage, then the MOSFET pinout
One motor only at full speed	Duty voltage COM_A1 or COM_B1 is outside the triangle range, so lower the PID or base gain
Random jitter at high speed	Add the filter capacitor to D, and check ripple on +5 V

Test log

Item	Expected	Measured	Pass
12 V current	80 - 120 mA		
+5 V	4.9 - 5.1 V		
VREF	2.45 - 2.55 V		
Triangle frequency	4.5 - 5.2 kHz		
VBASE range	1.2 - 3.8 V		
Duty at VBASE 2.5 V	50% on both		
S1 to S10 on white	0.3 - 1.5 V		
S1 to S10 on black	3.5 - 4.9 V		
VERROR minus VREF after offset	under 5 mV		
VERROR with line under S1	1.5 - 2 V from 2.5 V		
VERROR with line under S10	equal and opposite		
Motor direction check	pass		